Physics of life

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Introduction



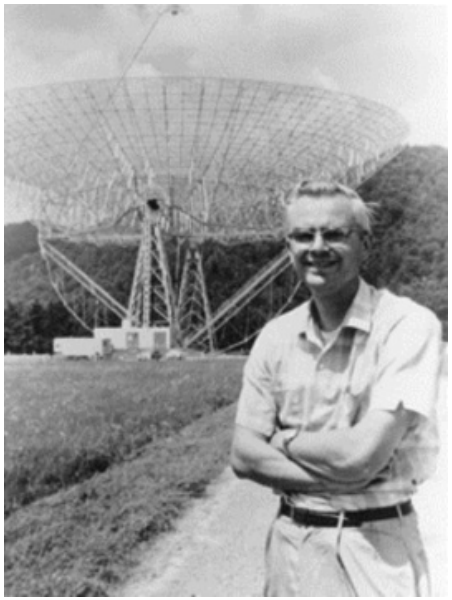
- Is there life outside the solar system?
- How do we make the question scientific?

Assessment

- This topic can be part of your presentation.
- Additional topics possible, such as the Physics of Wind Farms on Mars.

SETI

SETI, the Search for Extraterrestrial Intelligence, is an exploratory science that seeks evidence of life in the universe by looking for some signature of its technology.



- Search for electromagnetic radiation from Alien civilisations.
- The Drake Equation was developed by Frank Drake in 1961 as a way to focus on the factors which determine how many intelligent, communicating civilizations there are.

The Drake equation

$$N = R * f_p * n_e * f_l * f_i * f_c * L$$

N = The number of civilizations in The Milky Way Galaxy whose electromagnetic emissions are detectable.

R = The rate of formation of stars suitable for the development of intelligent life.

f_p = The fraction of those stars with planetary systems.

n_e = The number of planets, per solar system, with an environment suitable for life.

f_l = The fraction of suitable planets on which life actually appears.

f_i = The fraction of life bearing planets on which intelligent life emerges.

f_c = The fraction of civilizations that develop a technology that releases detectable signs of their existence into space.

L = The length of time such civilizations release detectable signals into space.

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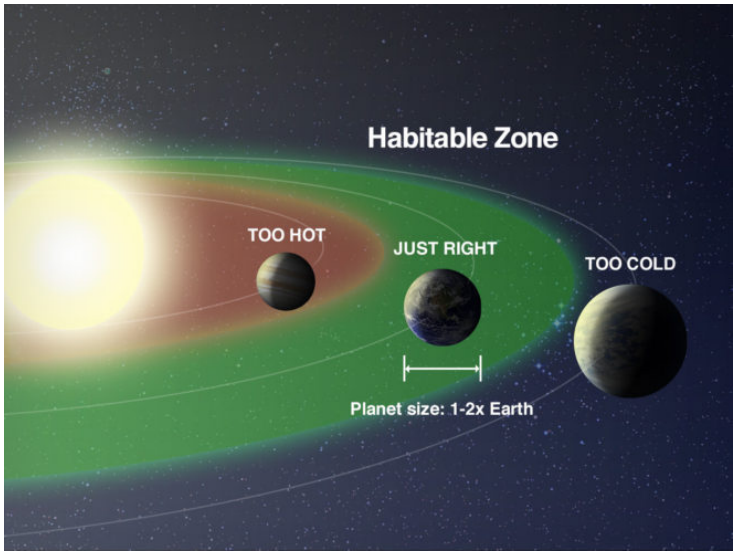
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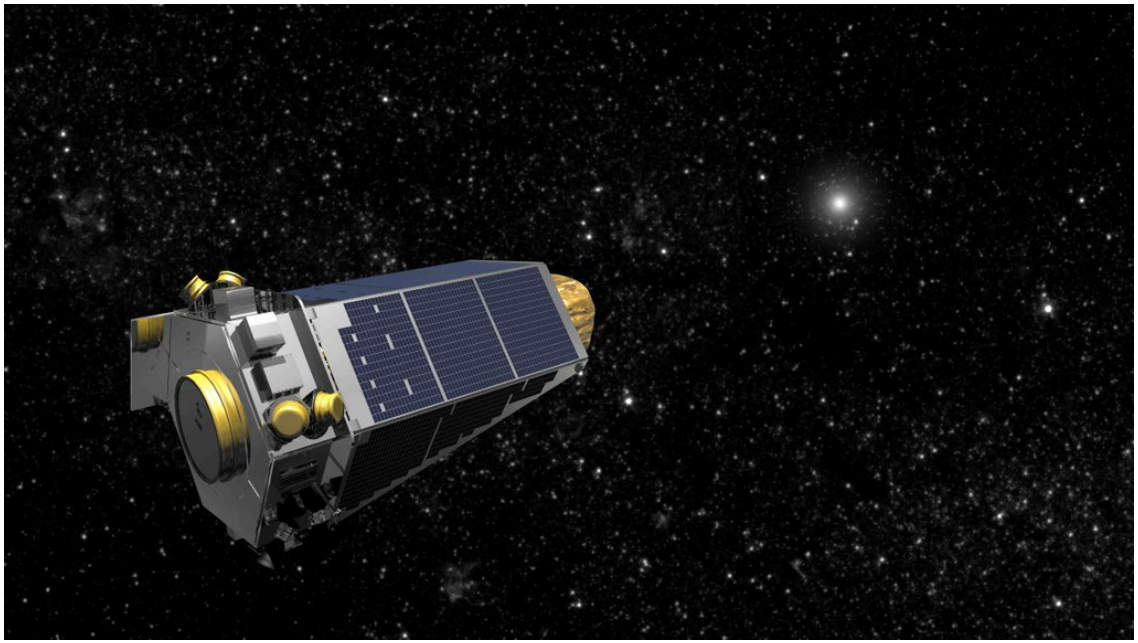
Goldilocks zone

- The Goldilocks Zone is the relatively narrow range of distance from a star within which an orbiting planet may be able to sustain life, because the temperature is right for liquid water.

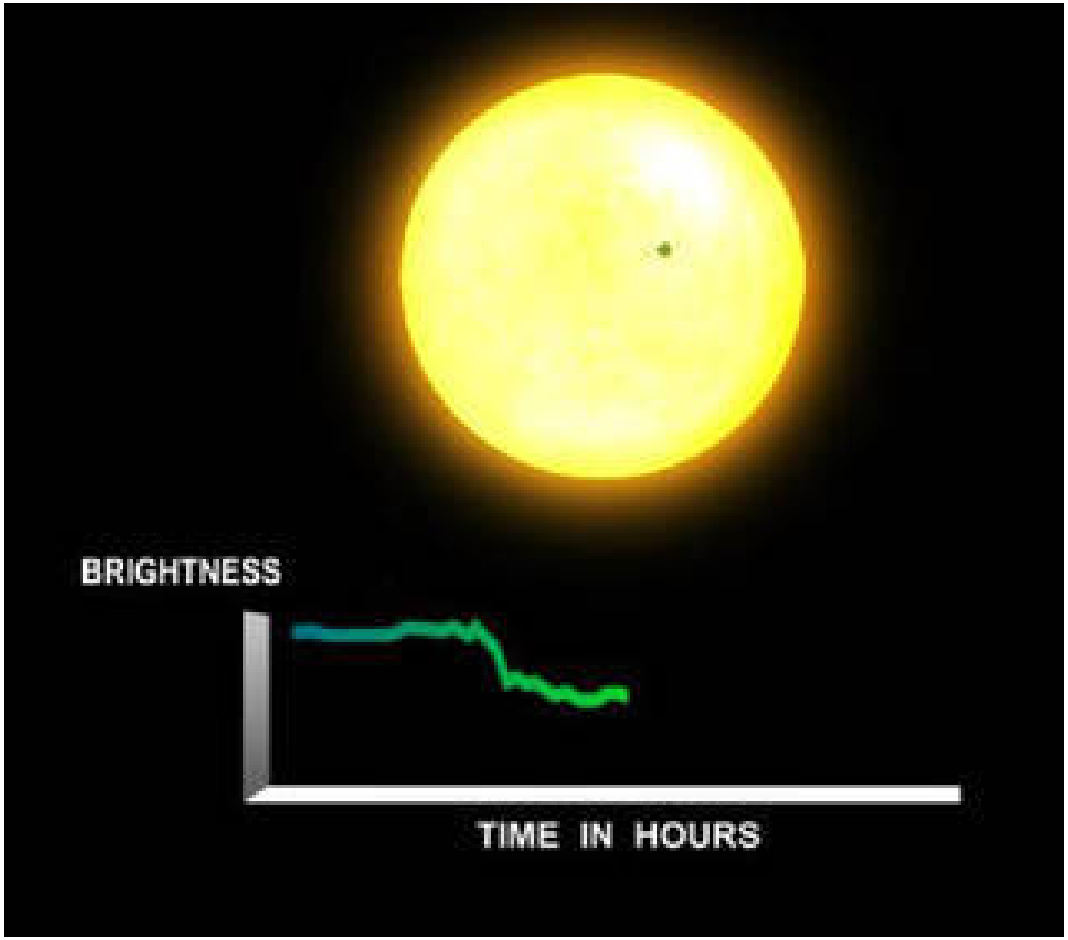


The Kepler mission

- The Kepler space observatory was launched in 2009.
- Kepler's only scientific instrument is a photometer.



The transit method



Summary of exoplanets

Source from NASA

- Candidate exoplanets: 4,696
- Confirmed exoplanets: 2,331
- Confirmed exoplanets less than twice Earth-size in the habitable zone: 21

Drake's equation

- As many skeptics have pointed out, the Drake equation can give a very wide range of values, depending on the assumptions.
- N is The number of civilizations in The Milky Way Galaxy whose electromagnetic emissions are detectable.
- L = The length of time such civilizations release detectable signals into space.
- N = 36.4 million using $L = 10^9$ years
- N = 4×10^{-12} using L= 304 years.

Conclusions

- The development of life on Earth required a number of conditions to occur, such as the Earth being in the correct orbit, so that there are liquid oceans.
- Physics principles are important for life.

This lecture was part of a MATH1607PP, **The Quantum Universe** module, taught 2015 to 2018.
ASSESSED LEARNING OUTCOMES:

1. Produce an overview of phenomena at all scales of the universe and explain the corresponding astrophysical relevance.
2. Use the knowledge on quantum phenomena to explain modern technologies of the medical sciences
3. Present results professionally.